



COLLEGE
OF
MEDICINE

Leishmaniasis

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Leishmaniasis

Leishmania refers to intracellular protozoan parasites of vertebrates that are transmitted by sand fly bites, associated with a variety of zoonotic diseases, including skin, mucosal, and systemic infections



Taxonomy and Classification of Leishmania Spp

- The causative agents of leishmaniasis are protozoan parasites of the genus *Leishmania*, which are classified under
 - The family Trypanosomatidae, within the class
 - Kinetoplastida, and
 - The phylum Euglenozoa.
- Kingdom: Protista Phylum: Protozoa Subphylum: Sarcomastigophora Class: Zoomastigophora Order Kinetoplastida Family: Trypanosomatidae Genus: *Leishmania*



Clinical forms of leishmaniasis

There are three clinical forms of leishmaniasis: cutaneous, mucocutaneous, and visceral.

- 1- Species causing **visceral leishmaniasis** (Leishmania Donovanii, Leishmania Infantum, Leishmania Chagasi)
- 2- Species causing **mucocutaneous leishmaniasis** (Leishmania Braziliensis, L. Infantum, Leishmania Tropica Mexicana)
- 3- Species causing **cutaneous leishmaniasis** L. Tropica and L major.

In Asia and the Middle East, the disease continued to be documented, and conditions termed the "Aleppo boil," "Jericho boil," and the "Baghdad boil"



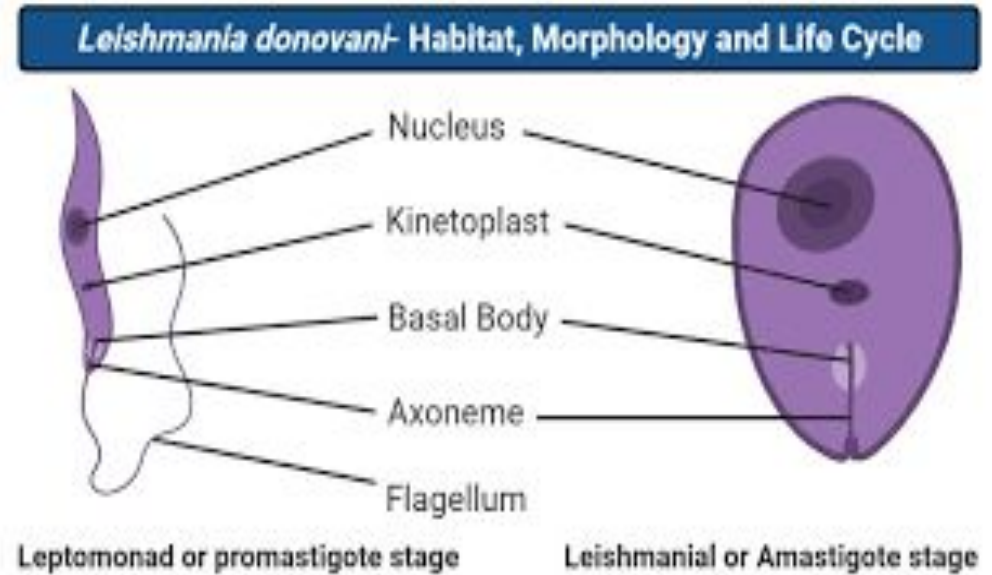
Leishmania classification

Categorization by geographic occurrence:

- In this system, disease is divided into :

- (1) Old World leishmaniasis (caused by *Leishmania* species found in Africa, Asia, the Middle East, the Mediterranean, and India), which produces cutaneous or visceral disease.
- (2) New World leishmaniasis (caused by *Leishmania* species found in Central and South America), which produces cutaneous, mucocutaneous, and visceral disease

Mode of transmission



- Leishmaniasis is transmitted by the bite of infected female phlebotomine sandflies.
- The sandflies inject the infective stage (i.e., promastigotes) from their proboscis during blood meals .



Life cycle:

A. The Vertebrate host:

When a sand fly holds the infective stage of Leishmania (the promastigote stage) takes a blood meal from man or other mammals, it will passively inject these parasites which will be picked up later by the reticuloendothelial system and transform to amastigote stage.

It is passing through sequence binary fissions until cells are filled with the parasites and burst to release these parasites which may infect other cells.

At times, many infected phagocytic cells arrive to circulation and to viscera, where the parasite settles down and multiply in the reticuloendothelial cells of the liver, spleen and bone marrow, causing the destruction of these cells. → which will cause signs & symptoms



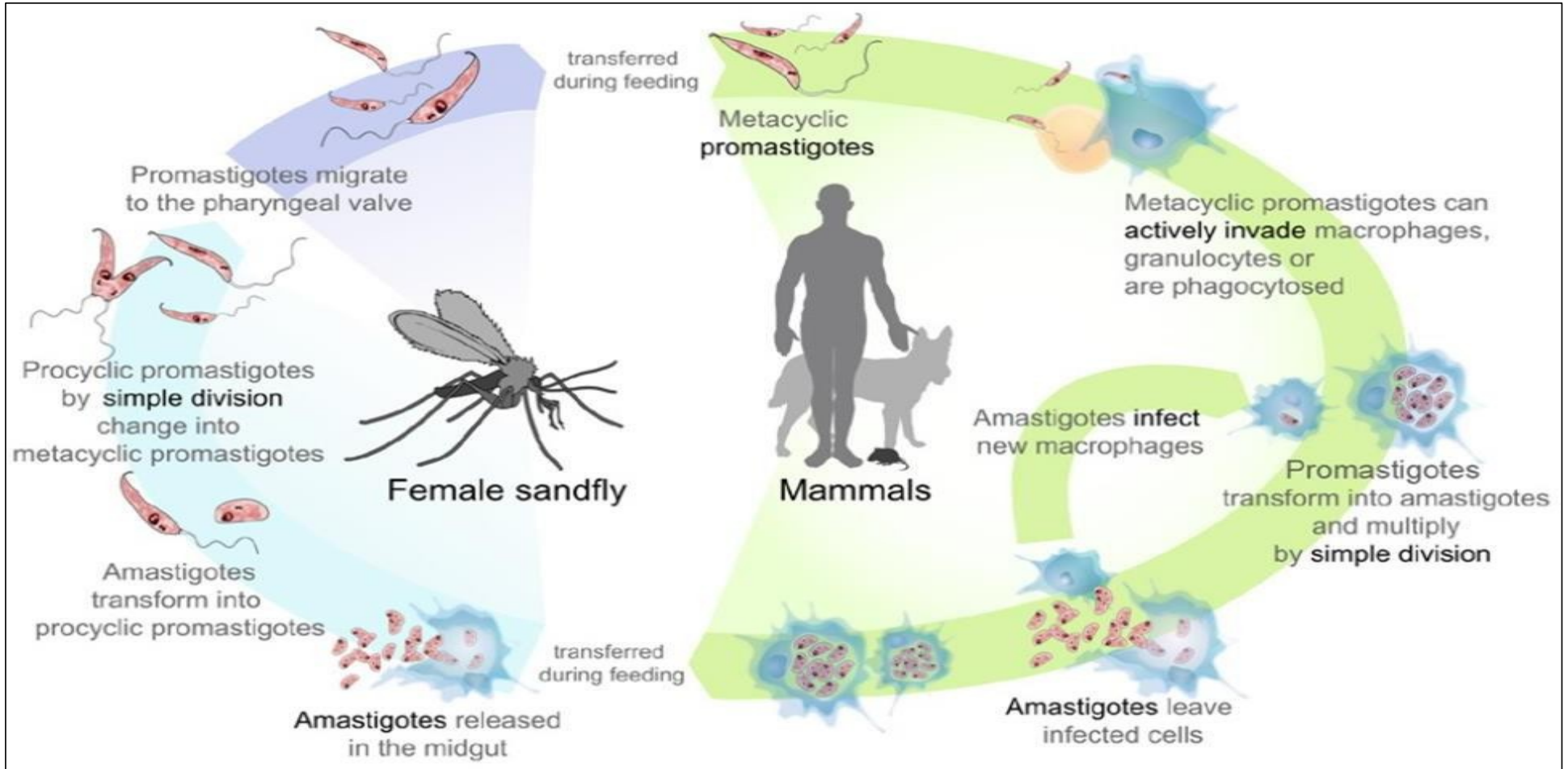
B. The Invertebrate host:

- These are many species of **sand fly** from genus *Phlebotomus*.
- However, when the insect bites the infected vertebrate host, it will suck the blood and the parasite (amastigote stage) which migrate then to the midgut of the insect and transform to promastigote stage and start in multiplication by binary fission.
- The parasites may be sticks on the intestinal wall of the insect or stay free in the lumen of the intestine. It may be found in aggregations of promastigote stage in the first and last part of the gut.

After 4 to 5 days from the insect feeding, the promastigotes (the infective stage) will fill the esophagus and when they block the esophagus, the insect will push the esophagus contents to the front and back; by this method it will inject the infective stages in the victim.



Lifecycle





Epidemiology of leishmaniasis

- The disease is distributed in 98 countries worldwide,
- An estimated 700 000 to 1.3 million new cases occur annually, with an about 20,000 to 40,000 deaths each year.
- The total incidence rate of cutaneous leishmaniasis in Iraq varies from 2.3 / 100,000 to 45.5 / 100,000 (
- Cutaneous leishmaniasis is prevalent across the whole of Iraq except for the 3 Kurdistan region governorates.
- in 2022, Iraq records 8000 cases of cutaneous leishmaniasis and estimated around 1800 visceral cases annually, with estimates of 400–5000 cases in previous years .



Epidemiology

- **Visceral leishmaniasis:** Most cases occur in Brazil, east Africa and India. An estimated 50 000 to 90 000 new cases of VL occur worldwide annually, with only 25–45% reported to WHO. It has outbreak and mortality potential.
- **Cutaneous leishmaniasis :** About 95% of CL cases occur in the Americas, the Mediterranean basin, the Middle East and central Asia. It is estimated that 600 000 to 1 million new cases occur worldwide annually but only around 200 000 are reported to WHO.
- **Mucocutaneous leishmaniasis :** Over 90% of mucocutaneous leishmaniasis cases occur in Bolivia (the Plurinational State of), Brazil, Ethiopia and Peru.



Cutaneous Leishmaniasis

- Pure Cutaneous Leishmaniasis was first described in the Old World by Lewis and Cunningham in 1876.
- It is caused by *L. tropica*. and *L. mexicana*. It occurs in areas of the body exposed to insect bites;
- in decreasing order of frequency, the most involved regions are the ears, nose, upper lip, cheeks, legs, hands and forearms, and ankles.) → exposed skin areas
- The incubation period is from 1 to 4 weeks but can last for up to several years. Patients may refer to previous travel to endemic zones.



Cutaneous Leishmaniasis

- The disease is known as **oriental sore** or **Baghdad boil** or **Old world cutaneous Leishmaniasis** or **tropica sore**. The vector is ***Phlebotomus papatasi*** and ***Phlebotomus sergenti***.
- The tissue reaction is initiated with the introduction of the promastigotes into the dermis.

Leishmania tropica

- The macrophages in the vicinity picks up the parasites which rapidly transform to amastigotes and multiply, destroying the host cells.
- Soon after, there will be a dense concentration of macrophages in the invaded area, all of which are liable to infection and destruction. *→ they come to help but they die*
- The lesion resulted from the insect bite become necrotic at the center, and the margins containing parasitized macrophages which may become infiltrated with giant and plasma cells.

Pathology and Symptoms

The lesion appears in the beginning as a **macula**, then as a **papule**, with a slightly raised center covered by a **thin blister like layer of epidermis**. Then it breaks down, with discharge of a small amount of clear or purulent exudates. *looks like ulcer*
↓
due to secondary bacterial infection



Pathology and Symptoms

The incubation period may be as short as 2 weeks or as long as 3 years, but usually between 2-6 months. In uncomplicated cases there are no systemic manifestations. The common occurrence of phonic complications causes painful, disfiguring, local ulcer, neutrophilic leukocytosis and fever, and at times septicemia.

if not treated it leads to heal with scars





Diagnosis

- The diagnosis of CL is based on *cutaneous leishmaniasis* **clinical features** (supported by **epidemiologic data**) and **laboratory testing**.
- The uncomplicated lesion may be mistaken for a variety of infections of the skin; hence demonstration of the parasite is essential. *→ especially bacterial*
- Examination of Giemsa stained slides of the relevant tissue is still the technique most commonly used to detect the parasite. As well as the serological methods.



Diagnosis

- Direct parasitological examination (microscopy, histopathology, and parasite culture). Parasitological confirmation is typically achieved through techniques such as lesion scarification, biopsy, imprint smears, or fine-needle aspiration—usually performed at the active margin of the lesion.
- These methods are rapid and cost-effective but have reduced sensitivity in chronic lesions.

Diagnosis

- Isolation of the organism in culture medium (using for example the diphasic NNN medium) or in experimental animals (hamsters) constitutes another method of parasitological confirmation of the diagnosis.
- Antibody detection can prove useful in visceral leishmaniasis but it is of limited value in cutaneous disease, where most patients do not develop a significant antibody response.
- Other diagnostic techniques including the isoenzymes, immunoassays and molecular approaches (PCR). *→ best way*



Leishmania donovani

- This parasite causes a disease called **Kala-azar** or **Dum-Dum fever** or **visceral Leishmaniasis** or **black fever**. Its vector is the **sand fly** (***Phlebotomus papatasi***).
- In the body of the vertebrate host, the parasite spreads in many parts, particularly, in the **endothelial cells** of the **blood and lymph capillaries of the spleen, liver and bone marrow**.
- Sometimes, the parasite infects the skin again and multiplies there. In this stage, the disease called **dermal Leishmaniasis** or **post Kala-azar**.

The incubation period varies from 10 days to many months, but in the usual case, it is insidious. On the average about 90 days following exposure. The symptoms can be observed are:

1. Great enlargement in the spleen and the liver with an increase in the size and number of parasitized Kupffer cells.
2. The bone marrow exhibits markedly increased production of macrophages and decreased erythropoietic function.
↳ RBC, ↓ Platelet, ↓ WBC (beside macrophages) production
3. Thrombocytopenia results in multiple hemorrhages, particularly from the mucous membranes.



In the typical acute case:

1. Temperature fluctuates daily from 37.5 - 40°C.
2. Bleeding ^{→ due to thrombocytopenia} typically occurs from the gum, lips, naris and intestinal mucosa.
3. Complications usually observed in **kala-azar** are principally diarrhea or dysentery and broncho-pneumonia.



Diagnosis of visceral leishmaniasis

- The parasite detection in tissue samples often necessitates invasive procedures, which limits their routine use. *since not in skin → in liver, spleen, bone marrow*
- Bone marrow aspiration is the most commonly employed technique, with a sensitivity ranging from 60% to 85%.
- Although splenic aspiration offers a higher sensitivity (over 95%), it is infrequently performed due to the significant risk of hemorrhage. *due to splenic rupture*
- Less invasive procedures, such as liver or lymph node aspiration, present lower sensitivities—approximately 45%.



Diagnosis of visceral leishmaniasis

- In vitro culture of clinical specimens or inoculation into laboratory animals can enhance diagnostic accuracy and increase positivity rates.
- However, these approaches are rarely utilized in clinical settings due to their technical complexity, time requirements, and limited practicality



Diagnosis of visceral leishmaniasis

- Polymerase chain reaction (PCR) has demonstrated considerable promise in the diagnosis of leishmaniasis. In addition to its high sensitivity, PCR can be applied to a variety of clinical specimens, including peripheral blood samples in cases of visceral leishmaniasis (VL).
- Moreover, certain PCR techniques allow for species-level identification of *Leishmania*, which is crucial for guiding appropriate treatment strategies.
- However, despite its advantages and widespread use in research settings, PCR is not routinely implemented in clinical diagnostics. *→ VL use clinical & epidemiological diagnoses, liver, Bone-marrow Biopsy*
- This is largely due to its high cost, the need for technical *proficiency*



Treatment

- The treatment of leishmaniasis depends on several factors including type of disease, concomitant pathologies, parasite species and geographic location.
- Leishmaniasis is a treatable and curable disease, which requires an immunocompetent system because medicines will not get rid of the parasite from the body, thus the risk of relapse if immunosuppression occurs.
- All patients diagnosed with visceral leishmaniasis *also for other types* require prompt and complete treatment.)



Treatment

- Treatment varies based on the form of leishmaniasis and the severity of the disease.
- Cutaneous Leishmaniasis (CL) is a Localized Disease. Often heals without treatment, but treatment can speed healing and reduce scarring.
- Treatment Options: Topical treatments like **paromomycin ointment**. Oral treatments like **miltefosine**.



Treatment

- Visceral Leishmaniasis (VL) first-line treatments
Amphotericin B (IV) is highly effective. Miltefosine (oral) as an alternative.
- Other options Sodium Stibogluconate (antimony-based), Paromomycin (injectable antibiotic)

not very safe → last resource in sure diagnosis

it not safe / safer fish



Treatment

- Mucocutaneous Leishmaniasis (MCL) it Treatment Systemic Antimonials (e.g., Sodium Stibogluconate).
- Amphotericin B for severe cases. Miltefosine for resistant cases.
- Post- Kala-Azar Dermal Leishmaniasis (PKDL) Treated similarly to VL, using amphotericin B, miltefosine, or antimonials.



Treatment

- Drug Resistance and Challenges: Resistance to antimonials is increasing, leading to the use of alternatives.
- Side effects and high costs of treatment, especially for amphotericin B, are challenges.



Treatment

- Supportive Care Management of symptoms like fever and nutritional support, particularly for VL. Wound care for CL lesions to avoid secondary infections. *→ if happen more risk of scarring and disfigurement*
- Prompt initiation of treatment is critical to preventing severe complications and limiting disease progression.
- The success of therapy depends on a combination of factors, including host-related elements such as genetic background, immune response, and the clinical manifestation of the disease.

Treatment

- Treatment-related factors such as drug quality, appropriate dosage, duration, and adherence to therapy also play a vital role.
- Additionally, characteristics of the parasite, including its inherent drug susceptibility and absence of resistance mechanisms, significantly influence treatment outcomes.
- Currently, pentavalent antimonial compounds remain the first-line pharmacological agents for the treatment of all clinical forms of leishmaniasis. *→ sodium - stibogluconate*



Prevention and control

Preventing and controlling the spread of leishmaniasis is complex and requires many tools. Key strategies include:

1- Early diagnosis and effective prompt treatment reduce the prevalence of the disease and prevents disabilities and death. It helps to reduce transmission and to monitor the spread and burden of disease.

2- Vector control helps to reduce or interrupt transmission of disease by decreasing the number of sandflies. Control methods include insecticide spray, use of insecticide-treated nets, environmental management and personal protection.



Prevention and control

3- Effective disease surveillance ^{→ know the number of cases} is important to promptly monitor and act during epidemics and situations with high case fatality rates under treatment.

4- Control of animal reservoir hosts is complex and should be tailored to the local situation.

5- Social mobilization and strengthening partnerships ^{→ people aware} — ^{→ intersectoral collaboration} mobilization and education of the community with effective behavioral change interventions must always be locally adapted.

* Personal protective measures like bed nets, long sleeves/pants, insect repellent for people living/travelling to endemic areas